

ouees-202606 topic 08:

AI or "Artificial Intelligence"

- What is it?**
- The philosophical implication**
- How it has changed the society**

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On the internet

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Lecture notes and reporting

- <https://github.com/jj1bdx/oueees-202606-public/>
- Check out the README.md file and the issues!
- Keyword at the end of the talk
- URL for submitting the report at the end of the talk

"Artificial Intelligence" (AI)

Fundamental question: *is intelligence only for human?*

What is intelligence?

Intelligence has been defined in many ways: the capacity for abstraction, logic, understanding, self-awareness, learning, emotional knowledge, reasoning, planning, creativity, critical thinking, and problem-solving.¹

¹ **Intelligence**, Wikipedia, (last visited June 21, 2026).

What is *artificial* intelligence?

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.²

² **Artificial intelligence**, Wikipedia, (last visited June 21, 2026).

AI as a buzzword in computer science (1/2) ³

- LISP and "symbolic" (text) processing (1950s)
- AI Winter in the 1970s
- Logic programming / expert systems (1980s)
 - Japan: The Fifth Generation Computer Systems (1982-1994)
- AI Winter in the 1990s

³ Wikipedia contributors, [History of artificial intelligence](#), Wikipedia, The Free Encyclopedia, 9 June 2023, 15:53 UTC [accessed 13 June 2023]

LISP language: code is data⁴

```
(defun add (a b) (+ a b))  
(format t "2 + 3 = ~a~%" (+ 2 3))  
(format t "cdr add = ~a~%"  
  (cdr (quote (defun add (a b) (+ a b))))))
```

2 + 3 = 5

cdr add = (ADD (A B) (+ A B))

⁴ [Online Lisp Interpreter](#), running SBCL 2.0.0

AI as a buzzword in computer science (2/2)

- Breakthrough made by large-scale computing
 - Internet, massive parallelism, cloud computing
- Since 2010s: generative AI
 - Machine learning -> deep learning
 - Massive "big data" from the internet
 - Large language models (LLMs) (e.g., GPT-5.5)

Logical certainty .vs. probabilistic certainty

- **Determinism: all events within the universe can occur only in one possible way**⁵
- **Logical reasoning: deduction or non-deductive ones (induction / abduction)**
- **Traditional algorithms assume determinism and deduction; obviously idempotent**
- **AI introduces *statistical* reasoning and *learning* algorithms**

⁵ **Determinism**, Wikipedia (last visited June 21, 2026).

How probability is misunderstood

- When the probability of an event is high, that does *not necessarily* mean the event will *always* happen
- Expected values do *not necessarily* represent the risk; *outliers matter*
- Normal distribution does *not necessarily* describe the whole set of actual samples

AI needs data; how are they collected? (1/2) ⁶

They are all collected *via the internet*

- Public web data: articles, blog posts, digitized books, academic papers, and forums**
- Source code: Massive repositories of public and open-source code, technical documentation, and scripts**
- Visual data: Billions of images, photos, artwork, diagrams, and charts**

⁶ **A conversation record between Kenji Rikitake and Gemini 3.5 Thinking**

AI needs data; how are they collected? (2/2)

- **Audio files: podcasts and speech recordings**
- **Videos: video frames and transcripts, such as filtered YouTube content**
- **Synthetic & human-curated alignment data: models undergo specialized training using Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF)**

**Question: can you respect
and retain copyright /
intellectual property
rights, privacy and
security of these collected
data and the owners?**

AI ignores the origins of data and questions

- You can prompt AI of revealing sensitive data
- You cannot protect any specific data in AI models from *unwanted* revelation
- AI is an algorithm and will not *proactively* act for protecting *sensitive* data

Can AI be legally creative? Unfortunately no

- **AI doesn't really care about plagiarism**
- **AI can generate wrong output quite often**
- **AI won't be able to handle copyright details**

Half of music professional refuse AI-generated music ⁷

Survey of 144 music supervisors, filmmakers, and advertisers by leading music companies

- 97% want to know if music is AI-generated**
- 49% will only work with human-made tracks**
- 40% said cultural authenticity becomes a critical selection factor**

⁷ **"Industry survey reveals 97% of music professionals demand AI transparency, while half refuse AI-generated music entirely"**, Record of the Day, 02 Dec 2025

Revisiting history of sampling music ⁸

- Modern hip-hop and many other pop music consist of samples from other people's pieces of works (drum breaks, impressive phrases, voices, etc.)
- Sampling technology emerged since 1980s, now pervasive everywhere even on your smartphone ⁹
- *The copyright issues are still in huge controversy*

⁸ **Sampling (music)**&oldid=1359206729), Wikipedia (last visited June 21, 2026).

⁹ For example, **Koala Sampler**

Resolving legal issues caused by AI-generated music and other forms of art might take many years

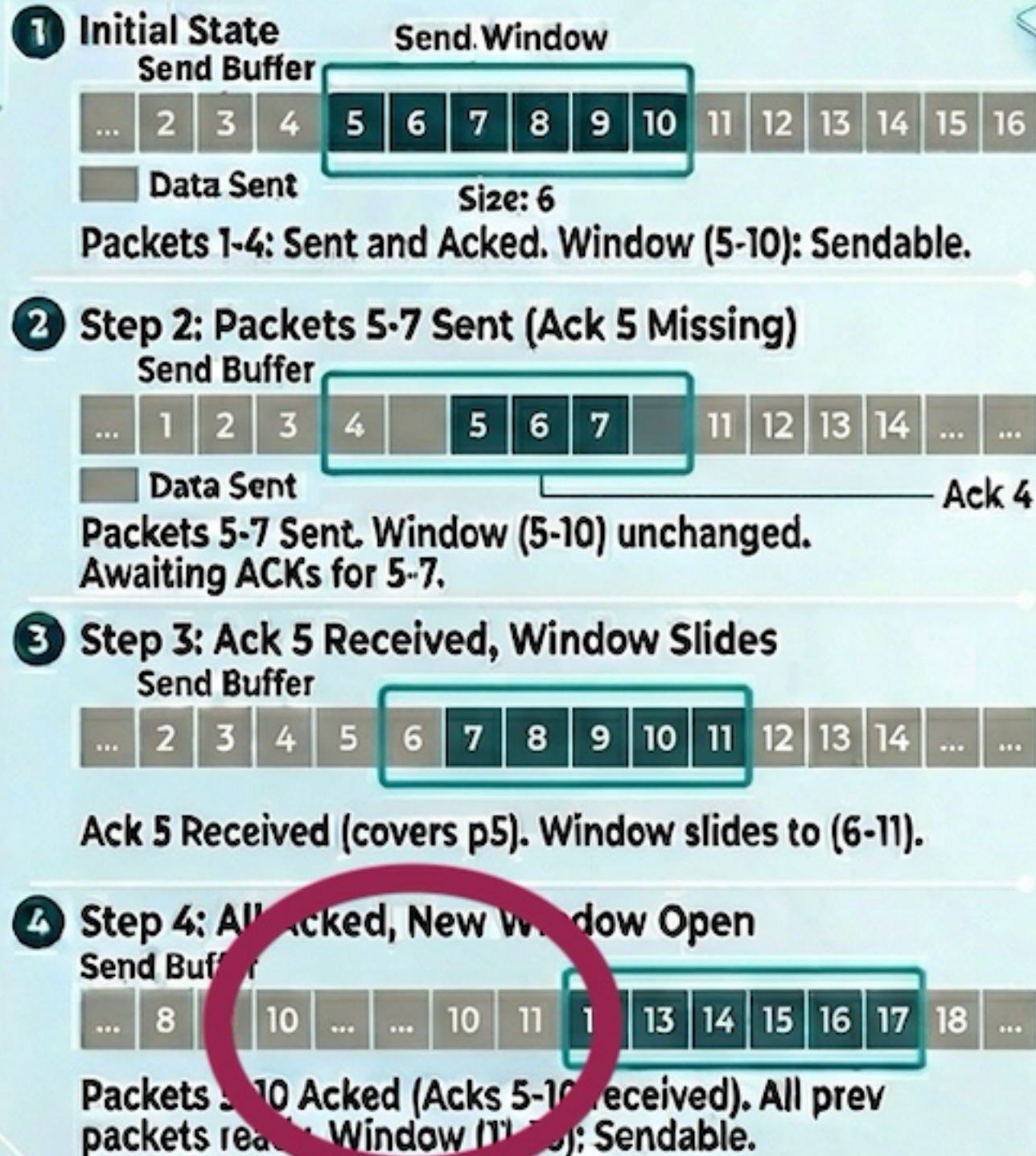
- AI output can be an ill-formed sampling
- AI-generated output is quite often inconsistent
- AI-generated output is derived from other people's data; identifying the source and owners is extremely difficult and virtually impossible

The (non-)goals of current AI

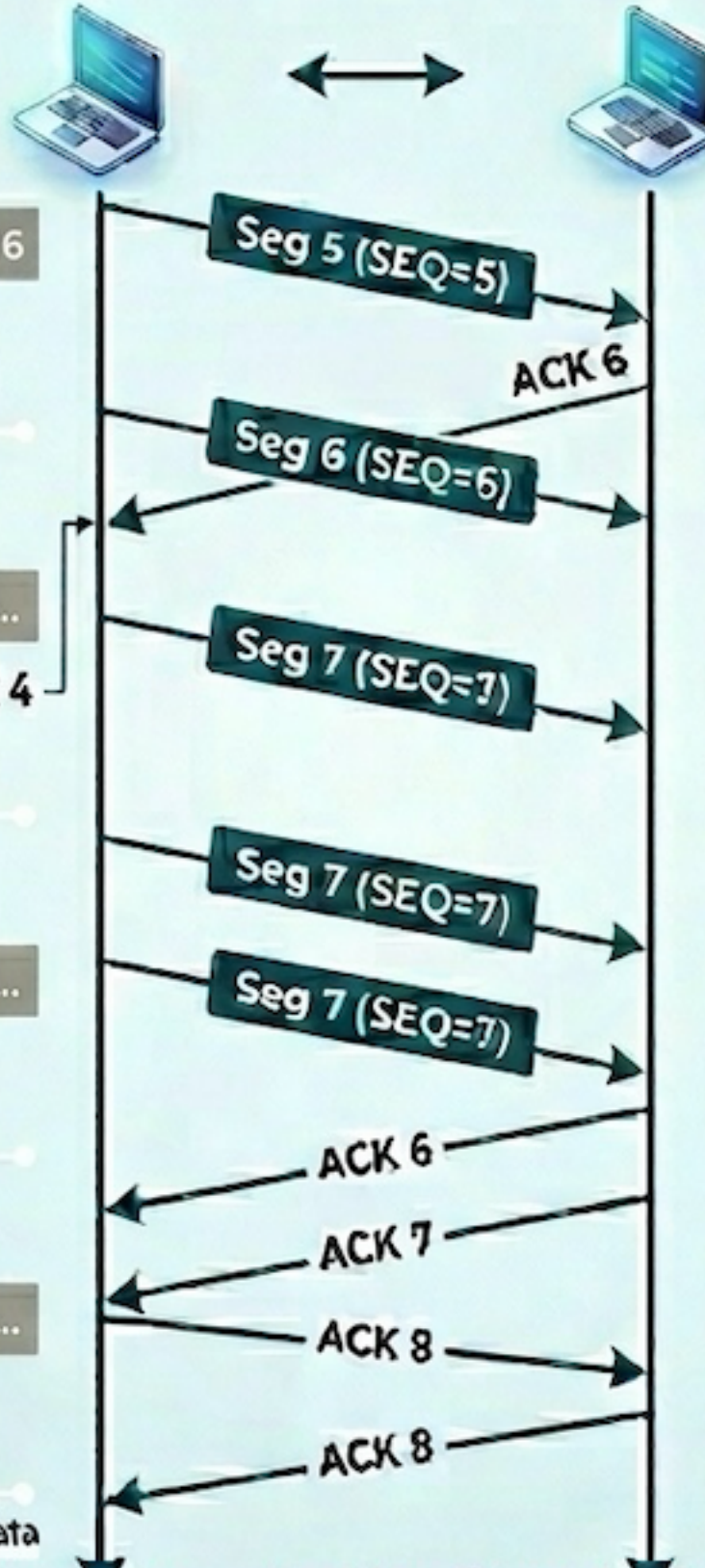
- ***Non-deterministic*** problem solving
 - Probabilistic answers computed by neural networks
 - Questionable reproducibility
 - ***Errors may quite often occur***
- ***No one really knows why*** LLMs work and what they actually do
 - You cannot ***fully logically*** explain the behavior

TCP SLIDING WINDOW PROTOCOL: A STEP-BY-STEP DIAGRAM

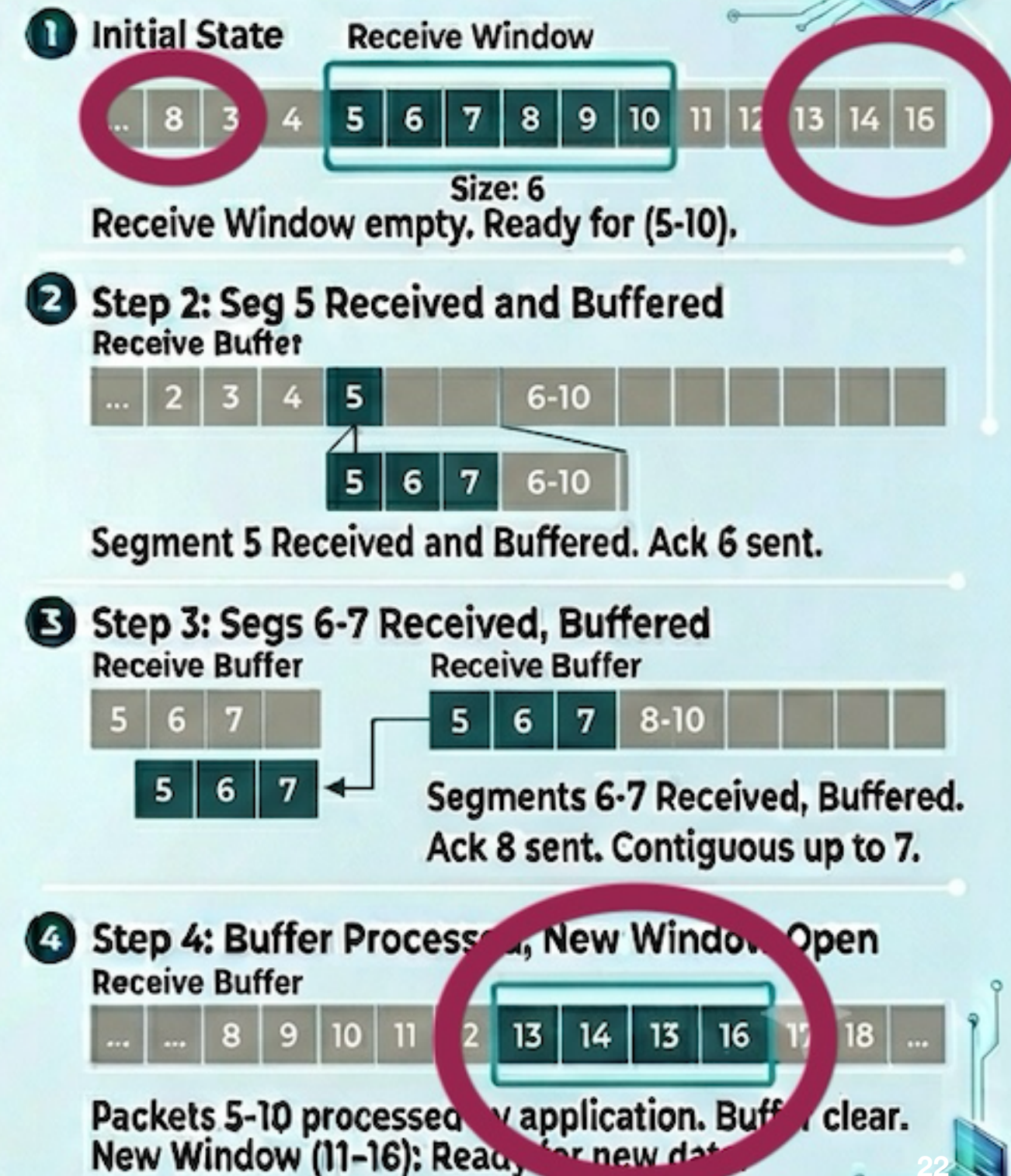
SENDER (HOST A)



Grey Bar = Data Aacked
Box = Sliding Window (Size: 6)
= Sent, Not Acked / Acceptable Data
= Out-of-Window / Unsendable



RECEIVER (HOST B)



Generative AI can only *estimate* answers

- Don't: expect the definitive answers
- Do: treat the output as the *synthesized* summary
 - The behavior is limited by the *given data*
 - Prompting (directives by words) largely affects the output
- Do: review the output by yourself
- *Don't: blindly believe the answers!*
- *Also don't: use the output without proper citation*

Generative AI (AI) raises fundamental and philosophical questions on making decisions

- AI outputs are probabilistic at best and is *not 100% deterministic*
- Modern science relies on deterministic logic for precision; should we give up precision for fast estimation using AI?
- Consequently, AI will induce more distrust against science

Social dehumanization and failures by AI

- Klarna hired back laid off 700 agents, once replaced with AI, failed to address the complex 20% of interactions ¹⁰
- AI-assisted software development puts huge burden on human developers ¹¹
- "[...] even if people think AI recommendations are low quality or not important, their decisions are still vulnerable to AI bias under certain circumstances." ¹²

¹⁰ Ashutosh Singhal, **Klarna Replaced 700 Agents with AI. Then Hired Them All Back.**, May 2, 2026

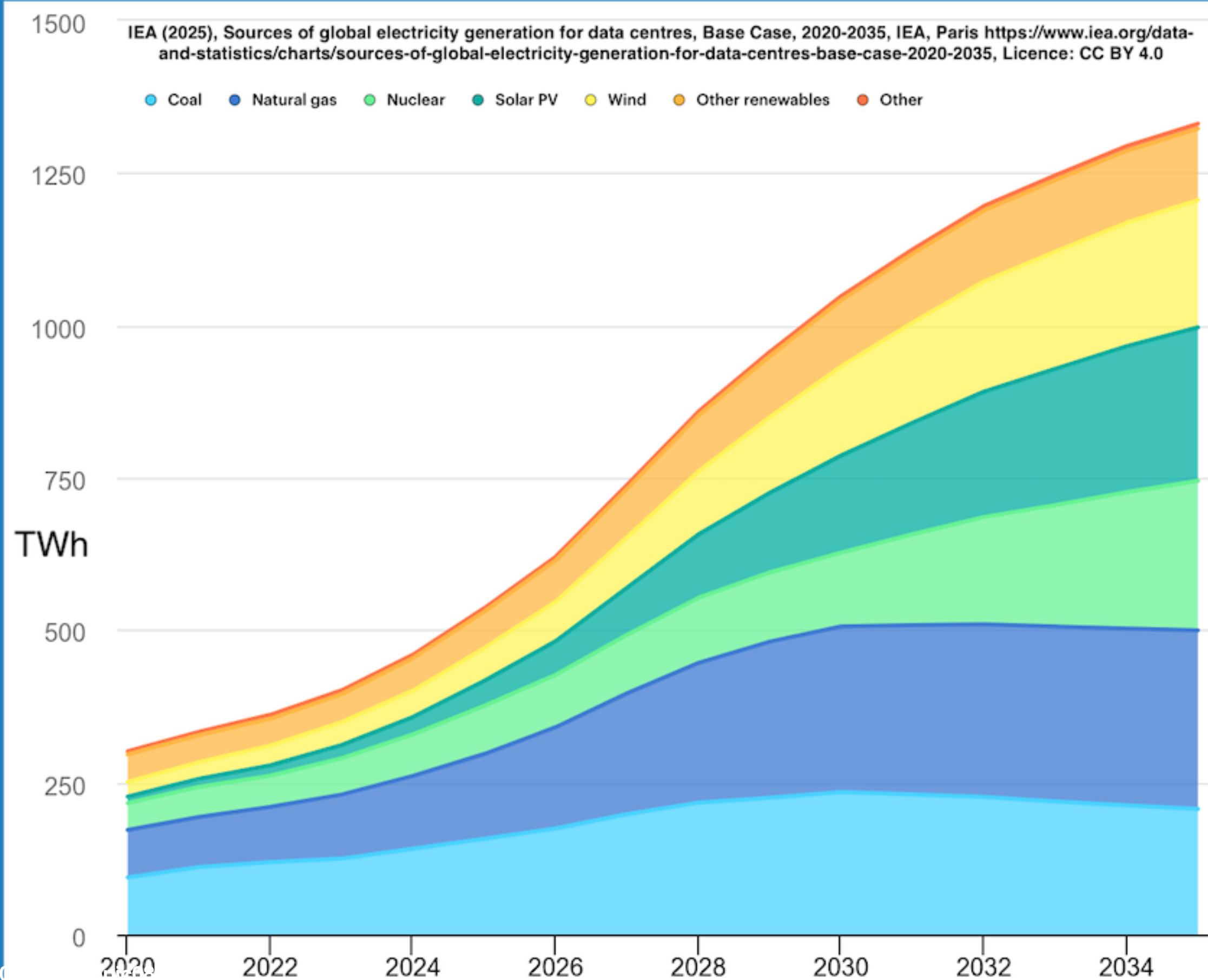
¹¹ S. Baltes, M. Cheong and C. Treude, **"An Endless Stream of AI Slop": How Developers Discuss the Burden of AI-Assisted Software Development**

¹² Wilson, K., Sim, M., Gueorguieva, A.-M., & Caliskan, A. (2025). No Thoughts Just AI: Biased LLM Hiring Recommendations Alter Human Decision Making and Limit Human Autonomy. Proceedings of the AAAI ACM Conference on AI, Ethics, and Society, 8(3), 2692–2704. <https://doi.org/10.1609/aies.v8i3.36749>

Environmental implication of AI (*Electricity alone*)

"Global electricity generation to supply data centres is projected to grow from 460 TWh in 2024 to over 1 000 TWh in 2030 and 1 300 TWh in 2035 in the Base Case. Over the next five years, renewables meet nearly half of the additional demand, followed by natural gas and coal, with nuclear starting to play an increasingly important role towards the end of this decade and beyond."

— IEA (2025), Energy and AI, IEA, Paris <https://www.iea.org/reports/energy-and-ai>, Licence: CC BY 4.0



Is investment to AI worth it? Does it enough good to the society equivalent to how people have paid for it?

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